Econometrics: worked problem sets

Linear regression, GLS, IV/GMM, likelihood and panel models

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1 Problem set 1

1.1 Question 1: OLS expectation and covariance

Text. Consider the linear regression model $y = X\beta + u$. Suppose that all the classical assumptions hold, in particular $u \sim N(0, \sigma^2 I)$ and $\mathbb{E}(u \mid X) = 0$.

(a) Show that the OLS estimator b is unbiased: $\mathbb{E}(b) = \beta$.

(b) Derive the conditional variance of b.

(c) How can we estimate the variance of the OLS estimator derived in (b)?

Complete solution. Let X be $n \times K$, of full column rank, and $n > K$. For a random design we use the classical conditional Gaussian assumption $u \mid X \sim N(0, \sigma^2 I_n)$; the unconditional Gaussian statement alone would not imply it. Define $A = (X'X)^{-1}X'$. The normal equations $X'(y - Xb) = 0$ have the unique solution

$$b = (X'X)^{-1}X'y = \beta + Au.$$

(a) Conditional on X, A is fixed. Consequently

$$\mathbb{E}[b \mid X] = \beta + A\mathbb{E}[u \mid X] = \beta.$$

Taking expectations again proves $\mathbb{E}b = \beta$, provided the estimator has finite expectation. Thus unbiasedness does not need Gaussian errors; the zero conditional mean and rank assumption suffice.

(b) Subtract the conditional mean and use the covariance rule for a linear transformation:

$$\begin{aligned}\operatorname{Var}(b \mid X) &= \mathbb{E}[Auu'A' \mid X] = A(\sigma^2 I_n)A' \\ &= \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}.\end{aligned}$$

In the Gaussian case, the complete conditional sampling law is $b \mid X \sim N(\beta, \sigma^2(X'X)^{-1})$.

(c) Put $P_X = X(X'X)^{-1}X'$ and $M_X = I_n - P_X$. The residual is $\hat{u} = M_X y = M_X u$, so the residual sum of squares is $RSS = u'M_X u$. The symmetric idempotent matrix M_X has rank and trace $n - K$. The identity $\mathbb{E}[u'Bu \mid X] = \operatorname{tr}(B \operatorname{Var}(u \mid X))$ for centered errors therefore gives

$$\mathbb{E}[RSS \mid X] = \sigma^2 \operatorname{tr}(M_X) = \sigma^2(n - K).$$

An unbiased estimator of the error variance is $s^2 = RSS/(n - K)$. Substitute it into the covariance formula:

$$\widehat{\operatorname{Var}}(b \mid X) = s^2(X'X)^{-1}, \qquad SE(b_j) = s\sqrt{[(X'X)^{-1}]_{jj}}.$$

Dividing RSS by n instead would underestimate σ^2 on average. Under conditional normality, $RSS/\sigma^2 \sim \chi^2_{n-K}$ independently of b, which supplies the exact t and F laws for inference.

1.2 Question 2: Coefficient of determination

Text. Consider the classical linear regression model $y = X\beta + u$.

(a) Derive the coefficient of determination, assuming the regression includes a constant, and discuss why this hypothesis is important.

(b) Discuss potential problems with the coefficient of determination and how they can be addressed.

Complete solution.

(a) Write $y = \hat{y} + \hat{u}$ with $\hat{y} = Xb$. OLS orthogonality gives $X'\hat{u} = 0$ and hence $\hat{y}'\hat{u} = 0$. If X includes the constant vector, it also gives $\sum_i \hat{u}_i = 0$, so the fitted mean equals $\bar{y}$. Therefore

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{u}_i.$$

Squaring and summing, the cross term vanishes because $\sum_i (\hat{y}_i - \bar{y})\hat{u}_i = \hat{y}'\hat{u} - \bar{y}\sum_i \hat{u}_i = 0$. Define

$$TSS = \sum_i (y_i - \bar{y})^2, \quad ESS = \sum_i (\hat{y}_i - \bar{y})^2, \quad RSS = \sum_i \hat{u}_i^2.$$

Then $TSS = ESS + RSS$. For $TSS > 0$ the fraction of centered sample variation explained is

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}, \qquad 0 \le R^2 \le 1.$$

Without a constant, the residuals need not sum to zero and this centered decomposition may fail; the centered R^2 can even be negative. The alternative uncentered quantity $1 - RSS/\sum_i y_i^2$ has a different denominator and should not be compared with a centered R^2.

(b) Adding regressors cannot increase the minimum residual sum of squares: the old solution remains feasible by assigning zero to the added coefficients. Hence ordinary R^2 never falls in nested models, even when useless variables are added. The adjusted measure

$$\bar{R}^2 = 1 - \frac{RSS/(n-K)}{TSS/(n-1)}$$

penalizes the loss of residual degrees of freedom and may fall after adding a regressor. Model comparisons require the same dependent variable and sample: a log-wage R^2 is not directly comparable to a wage-level R^2. A high R^2 does not establish exogeneity, causality or a correct functional form; a low value does not invalidate an precisely estimated coefficient. Use coefficient tests and residual diagnostics for inference, and held-out prediction error or justified information criteria for model selection. These assess questions that in-sample R^2 cannot answer.

1.3 Question 3: An intercept-only regression

Text. Consider the linear regression model $y_i = \beta_0$. Derive the least-squares estimator of the parameter and discuss the result.

Complete solution. The least-squares objective is $S(b_0) = \sum_{i=1}^n (y_i - b_0)^2$. Differentiate:

$$S'(b_0) = -2\sum_i (y_i - b_0) = -2\sum_i y_i + 2nb_0, \qquad S''(b_0) = 2n > 0.$$

The unique minimum solves $nb_0 = \sum_i y_i$, hence $b_0 = \bar{y}$. All fitted values equal the sample mean and $\sum_i (y_i - b_0) = 0$. When the equation is interpreted statistically as $y_i = \beta_0 + u_i$, with independent centered errors of variance σ^2, $\mathbb{E}\bar{y} = \beta_0$ and $\mathrm{Var}(\bar{y}) = \sigma^2/n$. If the displayed equation is literally deterministic with no error, every y_i equals β_0 and the same estimator recovers it exactly.

1.4 Question 4: Testing a subset of coefficients

Text. Consider $y = X\beta + u$ under the classical conditional Gaussian assumptions. Let J be the number of regressors whose coefficients are jointly tested against zero.

(a) Write a test using the residual sums of squares of the general and restricted models. How is the statistic distributed?

(b) Show how the test can be written using their R^2 values.

Complete solution.

(a) The unrestricted model has K coefficients including any intercept; the restricted model omits the J tested regressors. Denote their residual sums by RSS_U and RSS_R. Since the restricted parameter space is nested in the unrestricted one, $RSS_R \geq RSS_U$. Test H_0 that all J coefficients vanish, against the alternative that at least one does not. Under H_0,

$$\frac{RSS_R - RSS_U}{\sigma^2} \sim \chi^2_J, \qquad \frac{RSS_U}{\sigma^2} \sim \chi^2_{n-K},$$

and these quadratic forms are independent. To see why, the difference of the two residual projection matrices is an orthogonal projection onto the J newly fitted directions, orthogonal to the unrestricted residual space; Gaussian projections onto orthogonal subspaces are independent. Their normalized ratio is

$$F = \frac{(RSS_R - RSS_U)/J}{RSS_U/(n-K)} \sim F_{J,n-K}.$$

At significance level α, reject when $F > F_{J,n-K;1-\alpha}$. The exact law requires homoskedastic Gaussian errors; heteroskedasticity requires a robust covariance-based test.

(b) If both models contain a constant and use the same observations, they share TSS and $RSS_m = TSS(1 - R^2_m)$. Substitute into the preceding ratio:

$$F = \frac{(R^2_U - R^2_R)/J}{(1 - R^2_U)/(n-K)}.$$

The common TSS cancels. If only the intercept remains under H_0, then $R^2_R = 0$. Without a common sample and denominator this transformation is invalid.

1.5 Question 5: General linear restrictions

Text. In the same model, consider $H_0 : R\beta = r$, with R a $J \times K$ matrix and r a $J \times 1$ vector.

(a) Write an F statistic for these restrictions.

(b) For $K = 3$, write R and r to test jointly $\beta_2 = 3$ and $\beta_3 = 1 - \beta_1$.

Complete solution.

(a) Require R to have full row rank J. Under H_0, the vector $Rb - r$ has mean zero and covariance $\sigma^2 C$, where $C = R(X'X)^{-1}R'$ is positive definite. Whitening a Gaussian vector shows

$$\frac{(Rb - r)'C^{-1}(Rb - r)}{\sigma^2} \sim \chi^2_J.$$

Independence from $s^2 = RSS_U/(n-K)$ then gives

$$F = \frac{(Rb - r)'[R(X'X)^{-1}R']^{-1}(Rb - r)}{Js^2} \sim F_{J,n-K}.$$

Reject for a large upper-tail value. This formulation covers nonzero restrictions as well as excluded regressors. It equals the restricted/unrestricted residual-sum test because the least-squares restriction cost is exactly $(Rb - r)'C^{-1}(Rb - r)$.

(b) With $\beta = (\beta_1, \beta_2, \beta_3)'$, the second restriction is $\beta_1 + \beta_3 = 1$. Thus

$$R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad r = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad Rb - r = \begin{pmatrix} b_2 - 3 \\ b_1 + b_3 - 1 \end{pmatrix}.$$

The two rows are independent, so the preceding test uses $J = 2$ numerator degrees of freedom and $n - 3$ denominator degrees of freedom.

2 Problem set 2

2.1 Question 1: Quadratic wage-age regression

Text. The hourly wage rate in US dollars in the United States in 2019 is regressed on age, its square, a female dummy and a constant. The output is:

Source	SS	df	MS
Model	327210.935	3	109070.312
Residual	6916280.95	6066	1140.1716
Total	7243491.89	6069	1193.52313

$n = 6070$, $F(3, 6066) = 95.66$, $\Pr(> F) = 0.0000$, $R^2 = .0452$, $\bar{R}^2 = .0447$, root MSE $= 33.766$.

Variable	Coefficient	SE	t	p	95% CI	
age	2.209225	.2049644	10.78	.000	1.807422	2.611028
age2	-.0213555	.002253	-9.48	.000	-.0257722	-.0169389
female	-9.125352	.8675678	-10.52	.000	-10.82609	-7.424611
constant	-16.58503	4.384512	-3.78	.000	-25.18023	-7.989829

(a) What is the effect of increasing age by one year? Comment briefly.

(b) What is the predicted wage for a 40-year-old man?

(c) What can we say about the wage difference between men and women?

(d) Compute the statistic testing that all the explanatory-variable coefficients are zero, giving formula and result.

Complete solution. The fitted conditional mean is

$$\widehat{m}(a, f) = -16.58503 + 2.209225a - .0213555a^2 - 9.125352f.$$

(a) Its age derivative is $2.209225 - .042711a$, so the marginal effect is 1.355005 dollars at age 20 and $.500785$ dollars at age 40. The slope falls with age because the quadratic coefficient is negative. A literal one-year change is a discrete difference, not exactly the derivative:

$$\widehat{m}(a + 1, f) - \widehat{m}(a, f) = 2.209225 - .0213555(2a + 1).$$

It equals 1.3336495 at 20 and $.4794295$ at 40. Both answers distinguish the usual marginal interpretation from the finite one-year increment. The fitted curve reaches its maximum near $2.209225/.042711 = 51.72$ years, a model-based turning point rather than a causal conclusion.

(b) Substitute $a = 40$, $f = 0$, using the displayed coefficients without coarse rounding:

$$\widehat{m}(40, 0) = -16.58503 + 2.209225(40) - .0213555(1600) = 37.61517.$$

Thus the predicted hourly wage is about$ 37.62. Rounding the quadratic coefficient to $.021$ before multiplying by 1600 materially changes this calculation.

(c) At fixed age, switching f from zero to one changes the prediction by -9.125352 dollars per hour. This is a conditional difference; it is not the unconditional difference of the two sample means and is not automatically a causal effect. The reported standard error gives $t = -9.125352/.8675678 \approx -10.52$, so the conditional coefficient is statistically different from zero at conventional levels.

(d) Test $H_0 : \beta_{age} = \beta_{age^2} = \beta_{female} = 0$. The unrestricted model has $K = 4$ coefficients and the restricted model contains just a constant. Therefore

$$F = \frac{(RSS_R - RSS_U)/3}{RSS_U/6066} = \frac{327210.935/3}{6916280.95/6066} \approx 95.66.$$

Under the classical Gaussian assumptions its null law is $F_{3,6066}$; reject in the upper tail. The displayed p-value rounded to .0000 means very small, not literally zero. The alternative formula $[R^2/3]/[(1 - R^2)/6066]$ is equivalent, but using the rounded .0452 yields about 95.72 rather than the more precise table value. "All variables" here means the three slopes, as indicated by the output's three numerator degrees of freedom; including the intercept would be a different four-restriction test.

2.2 Question 2: Log wages and an interaction

Text. Here lrate is the natural logarithm of the hourly wage, age is measured in years, nchild is the number of children, female is a dummy, and nchild_female is their interaction.

Source	SS	df	MS
Model	166.618408	4	41.654602
Residual	3509.23844	6042	.580807421
Total	3675.85685	6046	.607981615

$n = 6047$, $F(4, 6042) = 71.72$, $\Pr(> F) = .0000$, $R^2 = .0453$, $\bar{R}^2 = .0447$, root MSE $= .76211$.

Variable	Coefficient	SE	t	p	95% CI	
age	.0072859	.0007604	9.58	.000	.0057952	.0087766
nchild	.0747049	.0114552	6.52	.000	.0522486	.0971611
female	-.1686385	.0245204	-6.88	.000	-.2167073	-.1205697
interaction	-.0820583	.016163	-5.08	.000	-.1137435	-.0503732
constant	2.856763	.0379752	75.23	.000	2.782318	2.931208

1) What is the estimated effect of age on the wage?

2) What is the estimated effect of having one child? Is it the same for men and women?

3) Compute the absolute log-wage difference between a man and a woman both aged 40 with one child.

4) Test significance of the interaction at $\alpha = .05$ with critical value 1.96. Give the null, alternative, statistic and law, rejection rule, conclusion and interpretation.

Complete solution. The fitted log wage is

$$\hat{\ell}(a, c, f) = 2.856763 + .0072859a + .0747049c - .1686385f - .0820583cf.$$

1) A one-year age increase changes the fitted log wage by .0072859. The first-order percentage approximation is .72859%, and the exact change in the exponentiated fitted log wage is $100(e^{.0072859} - 1) \approx .73125\%$. Exponentiating a conditional log mean gives a geometric-mean prediction; interpreting this as an arithmetic conditional mean requires an appropriate retransformation factor for the error.

2) For men ($f = 0$) an additional child changes log wage by .0747049: about 7.47049%, or exactly $100(e^{.0747049} - 1) \approx 7.7566\%$ in the exponentiated prediction. For women ($f = 1$), add the interaction coefficient:

$$.0747049 - .0820583 = -.0073534,$$

about $-.73534\%$, or exactly $100(e^{-.0073534} - 1) \approx -.73264\%$. The interaction means the child-related log-wage slope differs by $-.0820583$ between women and men. These estimates compare otherwise

identical covariate values; they do not establish a causal child penalty. Testing the female child slope itself would need the covariance between the two coefficient estimates, not supplied here.

3) For the same age and one child the common terms cancel:

$$\hat{\ell}(40,1,1) - \hat{\ell}(40,1,0) = -.1686385 - .0820583 = -.2506968.$$

Thus the absolute log-wage difference is .2506968. Its exponentiated interpretation is a female-to-male fitted ratio $e^{-.2506968} \approx .77826$, roughly 22.17% lower, rather than an exact 25.07% lower level wage.

4) Let δ denote the interaction coefficient. Test $H_0 : \delta = 0$ against $H_1 : \delta \neq 0$. The statistic is

$$t = \frac{\hat{\delta} - 0}{SE(\hat{\delta})} = \frac{-.0820583}{.016163} \approx -5.07692.$$

Under conditional Gaussian homoskedastic regression its exact null law is t_{6042}; the normal approximation uses the stipulated cutoff 1.96. Reject if $|t| > 1.96$; the result rejects H_0 strongly. There is evidence that the log-wage effect associated with an extra child differs by gender, with the female slope lower. This tests the difference in slopes, not the claim that every woman earns less than every man, nor the significance of either slope separately.

2.3 Question 3: Omitted-variable bias

Text. Assume the true model is $y = X\beta + Z\gamma + u$, but instead estimate $y = X\beta + u$. Is the OLS estimate unbiased for the true β? Assume Gaussian spherical errors and that all explanatory variables are exogenous.

Complete solution. Assume X has full column rank and $\mathbb{E}[u \mid X, Z] = 0$. The estimator computed in the omitted-variable regression is

$$b = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'Z\gamma + (X'X)^{-1}X'u.$$

Taking the conditional expectation gives

$$\mathbb{E}[b \mid X, Z] - \beta = (X'X)^{-1}X'Z\gamma.$$

Thus exogeneity of the included regressors with respect to u does not eliminate the omitted-variable term. The estimator is conditionally unbiased iff $X'Z\gamma = 0$, for instance if the omitted variables have zero coefficients or their relevant linear combination is sample-orthogonal to all columns of X. Unconditional unbiasedness instead requires the expectation of that whole design-dependent term to vanish. In a simple slope regression with an intercept, the population bias has the form $\gamma \operatorname{Cov}(x, z)/\operatorname{Var}(x)$. Hence its direction depends both on how the omitted variable affects y and on its association with x. The new composite error $Z\gamma + u$ is generally correlated with X, violating the condition needed for OLS consistency as well.

2.4 Question 4: Partitioned regression

Text. Suppose $y = X_1\beta_1 + X_2\beta_2 + \varepsilon$. What is the algebraic solution for b_2, and how can it be interpreted?

Complete solution. Assume X_1 has full column rank and the full matrix $[X_1, X_2]$ does too. The first block of the normal equations gives

$$b_1 = (X_1'X_1)^{-1}X_1'(y - X_2b_2).$$

Substitute this into the second block $X_2'(y - X_1b_1 - X_2b_2) = 0$. Defining $M_1 = I - X_1(X_1'X_1)^{-1}X_1'$, we obtain

$$X_2'M_1y - X_2'M_1X_2b_2 = 0, \qquad b_2 = (X_2'M_1X_2)^{-1}X_2'M_1y.$$

The inverse exists because the full design is of full rank. Now residualize both y and every column of X_2 on X_1: $\tilde{y} = M_1 y$, $\tilde{X}_2 = M_1 X_2$. Since M_1 is symmetric and idempotent,

$$(\tilde{X}_2' \tilde{X}_2)^{-1} \tilde{X}_2' \tilde{y} = (X_2' M_1 X_2)^{-1} X_2' M_1 y = b_2.$$

Thus the partial-regression coefficients come from regressing the part of y unexplained by X_1 on the part of X_2 unexplained by X_1. This is the Frisch–Waugh–Lovell identity; it explains algebraically what "holding the other regressors fixed" means.

2.5 Question 5: The simple-regression slope

Text. For $y_i = \beta_1 + x_i \beta_2 + \varepsilon_i$, use partitioned regression to derive b_2 and interpret it.

Complete solution. Set $X_1 = \mathbf{1}$ and $X_2 = x$. The residual-maker is $M_1 = I - n^{-1}\mathbf{1}\mathbf{1}'$, which subtracts sample means: $M_1 y$ has entries $y_i - \bar{y}$ and $M_1 x$ has entries $x_i - \bar{x}$. Consequently

$$b_2 = \frac{x' M_1 y}{x' M_1 x} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

The denominator must be positive, meaning x is not constant. The numerator and denominator are proportional to the sample covariance and sample variance with the same divisor, so $b_2 = s_{xy}/s_x^2$. It measures fitted change in y per one-unit change in x; it does not imply causality without exogeneity. The first normal equation supplies $b_1 = \bar{y} - b_2 \bar{x}$, so the fitted line passes through the point $(\bar{x}, \bar{y})$.

3 Problem set 3

3.1 Question 1: Nonspherical errors and GLS

Text. Consider $y = X\beta + u$ with $\mathbb{E}(u \mid X) = 0$ and $\mathrm{Var}(u \mid X) = \sigma^2\Omega$, where Ω is positive definite.

(a) Is OLS unbiased?

(b) Derive its covariance and compare with the homoskedastic case.

(c) If errors are heteroskedastic but uncorrelated, what is the form of Ω? Suggest a covariance estimator.

(d) For known Ω, derive an efficient estimator and its mean and variance.

Complete solution. Assume full column rank of the $n \times K$ matrix X throughout.

(a) OLS is $b = \beta + (X'X)^{-1}X'u$. Conditioning on X therefore gives $\mathbb{E}[b \mid X] = \beta$, irrespective of the shape of Ω. The estimator remains unbiased if its unconditional expectation exists. Nonspherical errors change precision, not this zero-mean calculation.

(b) The covariance rule for a linear transformation yields

$$\mathrm{Var}(b \mid X) = \sigma^2(X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

When $\Omega = I_n$, the central product is $X'X$ and the expression simplifies to $\sigma^2(X'X)^{-1}$. With a general Ω it need not simplify and can be larger in some directions and smaller in others; there is no universal ordering relative to a different homoskedastic model with the same scalar σ^2. OLS also ceases to be the best linear unbiased estimator unless the error structure happens to make OLS and GLS coincide.

(c) Uncorrelated errors give a diagonal matrix

$$\Omega = \mathrm{diag}(\omega_1, \ldots, \omega_n), \quad \omega_i > 0, \qquad \mathrm{Var}(u_i \mid X) = \sigma^2\omega_i.$$

Let $\hat{u}_i = y_i - x_i'b$. A heteroskedasticity-consistent sandwich estimate is

$$\widehat{V}_{HC0} = (X'X)^{-1}\left(\sum_{i=1}^{n} x_i x_i'\hat{u}_i^2\right)(X'X)^{-1}.$$

No separate estimate of σ^2 is needed because the squared residuals estimate the whole observation-specific variance. It is an asymptotically consistent covariance estimate under independent observations, adequate moments and no dominating leverage; it is not generally finite-sample unbiased. The degrees-of-freedom variant $HC1$ multiplies it by $n/(n-K)$. If errors are serially correlated too, off-diagonal covariance terms must be estimated as well, so this diagonal-error sandwich is insufficient.

(d) Positive definiteness permits whitening: choose $C = \Omega^{-1/2}$ with $C\Omega C' = I_n$ and transform $y^* = Cy$, $X^* = CX$, $u^* = Cu$. Then $\mathrm{Var}(u^* \mid X) = \sigma^2 I_n$, so OLS in the transformed model minimizes

$$(y - Xb)'\Omega^{-1}(y - Xb).$$

Differentiating with respect to b gives $-2X'\Omega^{-1}(y - Xb) = 0$, hence

$$b_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y.$$

Let $A_G = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}$. Since $A_G X = I_K$, $b_{GLS} = \beta + A_G u$ and $\mathbb{E}[b_{GLS} \mid X] = \beta$. Its covariance is

$$\mathrm{Var}(b_{GLS} \mid X) = \sigma^2 A_G \Omega A_G' = \sigma^2(X'\Omega^{-1}X)^{-1}.$$

To establish efficiency within linear unbiased estimators, let another estimator be Ay, with $AX = I_K$, and put $D = A - A_G$. Then $DX = 0$. The cross covariance is zero because $D\Omega A_G' = DX(X'\Omega^{-1}X)^{-1} = 0$. Therefore

$$\mathrm{Var}(Ay \mid X) - \mathrm{Var}(b_{GLS} \mid X) = \sigma^2 D\Omega D' \succeq 0.$$

GLS is thus best linear unbiased under the given covariance model. With conditionally Gaussian errors and known Ω, it is also the likelihood estimator of β. If Ω is unknown and replaced by an estimate, the resulting feasible GLS needs additional consistency conditions; exact finite-sample GLS properties do not transfer automatically.

3.2 Question 2: An instrument for one endogenous regressor

Text. In $y_i = x_{1i}'\beta_1 + x_{2i}\beta_2 + \varepsilon_i$, the vector x_{1i} contains $K-1$ exogenous variables, while x_{2i} is endogenous. An additional variable z_{2i} is observed.

(a) Write moment conditions using the exogenous variables and z_{2i}.

(b) Derive IV with $Z = [X_1, z_2]$ and $X = [X_1, x_2]$.

(c) State and discuss instrument validity.

(d) Discuss consistency and give the asymptotic distribution.

(e) Explain why IV standard errors are typically larger than corresponding OLS standard errors.

Complete solution.

(a) At the true parameter, the assumptions are

$$\mathbb{E}[x_{1i}\varepsilon_i] = 0_{K-1}, \qquad \mathbb{E}[z_{2i}\varepsilon_i] = 0, \qquad \mathbb{E}[z_i(y_i - x_i'\beta)] = 0_K,$$

where $z_i = (x_{1i}', z_{2i})'$. There is no required moment $\mathbb{E}[x_{2i}\varepsilon_i] = 0$: it is precisely the potentially violated OLS condition.

(b) The sample analogues are $Z'(y - Xb) = 0_K$. If $Z'X$ is invertible, solve them exactly:

$$b_{IV} = (Z'X)^{-1}Z'y = \beta + (Z'X)^{-1}Z'\varepsilon.$$

Here there are K moments and K coefficients, so the model is exactly identified. Sample invertibility alone is not enough for asymptotic identification; its population counterpart must be nonsingular too.

(c) Exogeneity requires $\mathbb{E}[z_{2i}\varepsilon_i] = 0$. Economically, the excluded instrument must not have an omitted direct effect on y or share unobserved determinants in ε. Relevance requires $Q_{zx} = \mathbb{E}[z_i x_i']$ to have full rank K. In the single endogenous-variable case, after linearly removing X_1 from both x_2 and z_2, their covariance must be nonzero. Mere correlation caused by common controls is insufficient. Exclusion and relevance are separate: significance in a first-stage regression says nothing by itself about exogeneity.

(d) Assume independent identically distributed observations, the moment conditions, sufficient finite second moments for the law of large numbers, and nonsingular Q_{zx}. Then

$$b_{IV} - \beta = (Z'X/n)^{-1}(Z'\varepsilon/n) \xrightarrow{p} Q_{zx}^{-1}0 = 0.$$

If additionally the multivariate central limit theorem applies to $z_i\varepsilon_i$, with $S = \mathbb{E}[z_i z_i'\varepsilon_i^2]$, Slutsky's theorem gives

$$\sqrt{n}(b_{IV} - \beta) \xrightarrow{d} N\left(0, Q_{zx}^{-1} S Q_{zx}^{-\prime}\right).$$

Under conditional homoskedasticity $\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2$, $S = \sigma^2 Q_{zz}$, $Q_{zz} = \mathbb{E}[z_i z_i']$. Estimate the robust version using IV residuals and $\hat{S} = n^{-1}\sum_i z_i z_i'\hat{\varepsilon}_i^2$, with an overall $1/n$ factor for the covariance of b_{IV} itself. Weak instruments, with relevance vanishing as n grows, invalidate this ordinary fixed-rank normal approximation.

(e) IV uses only the variation in x_2 predicted by the instruments after controls, rather than all its variation. Weak partial correlation produces little effective identifying variation and a large inverse Q_{zx}^{-1} in the covariance. Under homoskedasticity, valid OLS and a single endogenous-style regressor, the idealized variance inflation behaves like the reciprocal of the first-stage partial R^2. The comparison concerns precision, not overall validity: when x_2 is actually endogenous, smaller conventional OLS errors do not make its biased and inconsistent estimate preferable. A universal numerical ordering is not claimed for arbitrary heteroskedastic or differently specified models.

3.3 Question 3: Linear GMM

Text. Consider $R \geq K$ moments $\mathbb{E}[z_i \varepsilon_i] = 0$, with $\varepsilon_i = y_i - x_i'\beta$.

(a) Write the quadratic objective minimized in GMM.

(b) Derive its linear estimator.

(c) Write the optimal GMM estimator.

(d) Show weight independence when $R = K$.

(e) State when the estimator is OLS.

Complete solution. Let X be $n \times K$, Z be $n \times R$, and W_n be a symmetric positive-definite $R \times R$ weight matrix.

(a) The sample moment vector is $g_n(b) = Z'(y - Xb)/n$. Minimize

$$Q_n(b) = g_n(b)' W_n g_n(b).$$

Multiplying the objective by n or n^2 does not change its minimizer. An arbitrary fixed weight generally produces a consistent but not efficient GMM estimator under valid, identifying moments.

(b) Differentiating the objective gives

$$\nabla_b Q_n(b) = -\frac{2}{n^2} X'ZW_nZ'(y - Xb).$$

Setting it to zero and assuming $X'ZW_nZ'X$ invertible yields

$$b_W = (X'ZW_nZ'X)^{-1}X'ZW_nZ'y.$$

The Hessian is $2X'ZW_nZ'X/n^2$, positive definite under the rank assumption, so this stationary point is the unique global minimizer.

(c) Let S be the asymptotic covariance of $\sqrt{n}g_n(\beta)$. For independent observations $S = \mathbb{E}[z_i z_i' \varepsilon_i^2]$; dependence would instead require a long-run covariance. The efficient choice is $W = S^{-1}$. With a preliminary consistent estimate $\tilde{b}$, define $\hat{\varepsilon}_i = y_i - x_i'\tilde{b}$ and $\hat{S} = n^{-1}\sum_i z_i z_i' \hat{\varepsilon}_i^2$. The two-step feasible estimator is

$$b_{eff} = (X'Z\hat{S}^{-1}Z'X)^{-1}X'Z\hat{S}^{-1}Z'y.$$

For $G = \mathbb{E}[z_i x_i']$ of full column rank, its efficient asymptotic covariance is $(G'S^{-1}G)^{-1}$ for $\sqrt{n}(b_{eff} - \beta)$, or this divided by n for the coefficient estimator. Under homoskedasticity $S = \sigma^2 \mathbb{E}[z_i z_i']$; the scalar σ^2 cancels from the minimizer and the weight $W_n = (Z'Z)^{-1}$ yields the two-stage least-squares formula $(X'P_Z X)^{-1}X'P_Z y$, $P_Z = Z(Z'Z)^{-1}Z'$.

(d) If $R = K$ and $Z'X$ is invertible, there is a unique coefficient vector $b = (Z'X)^{-1}Z'y$ making $g_n(b) = 0$. The objective then equals zero, its smallest possible value for every positive-definite weight. Thus every such weight has the same minimizer. If the rank condition fails, this argument does not provide a unique identified estimator.

(e) If all regressors are exogenous and the instrument matrix is exactly $Z = X$, the moments are $X'(y - Xb) = 0$. The case $R = K$ just proved gives $b = (X'X)^{-1}X'y$, irrespective of weight. More generally, suitable choices of instruments and weights can yield the same projection, but exogeneity alone without specifying Z and W_n does not make every overidentified GMM estimator equal to OLS.

3.4 Question 4: The first stage in the education-earnings model

Text. The first stage of a 2SLS estimation of education's effect on earnings gives:

$n = 3010$, $F(6, 3003) = 608.02$, $\Pr(> F) = .0000$, $R^2 = .4745$, $\bar{R}^2 = .4734$, root MSE $= 1.9425$.

ed76 regressor	Coefficient	Robust SE	t	p	95% CI	
exp	-.4100081	.0319791	-12.82	.000	-.4727113	-.3473049
exp2	.0732287	.1702908	.43	.667	-.2606697	.4071272
black	-1.006138	.0877849	-11.46	.000	-1.178263	-.8340136
reg76r	-.291464	.0784454	-3.72	.000	-.4452761	-.1376519
smsa76r	.4038769	.0849467	4.75	.000	.2373173	.5704364
nearc4	.3373208	.0806044	4.18	.000	.1792753	.4953663
constant	16.65917	.1467408	113.53	.000	16.37145	16.9469

Education is ed76; exp is experience, exp2 is experience squared divided by100, black denotes race, reg76r residence in the South, smsa76r an urban area, and nearc4 whether the person grew up near a four-year college.

(a) Identify the endogenous variable, the excluded exogenous instrument and the other first-stage variables.

(b) How can we assess instrument weakness, and what does this output imply?

(c) Can we test whether the instrument is really exogenous, meaning it has no direct role in the earnings equation after the other covariates?

Complete solution.

(a) Education is treated as endogenous because unobserved ability or family background may affect both schooling and earnings. The excluded instrument is nearc4. The other listed regressors and the constant are included exogenous controls; they appear in both the earnings equation and the schooling first stage. Exclusion requires college proximity to affect earnings only through schooling after these controls; this is an identifying assumption, not an observed fact established by this output.

(b) Test the coefficient of the excluded instrument in the first stage, conditional on the included controls. With a single excluded instrument, the one-restriction Wald F is the square of its t statistic:

$$t = \frac{.3373208}{.0806044} \approx 4.18489, \qquad F_{excluded} \approx 17.5133.$$

The output's rounded $t = 4.18$ gives 17.4724. Both exceed the familiar heuristic cutoff 10, suggesting useful first-stage relevance. The overall 608.02 F includes all six slopes and is not the relevant excluded-instrument strength statistic. Moreover the reported covariance is robust: a mechanical cutoff 10 is only a heuristic and does not by itself guarantee accurate conventional IV inference. Weak-identification diagnostics and identification-robust inference may still be needed for a rigorous empirical analysis. This table establishes evidence of relevance, not validity.

(c) There is one endogenous variable and only one excluded instrument, so this is exactly identified. All K moment restrictions can be set to zero by the K coefficients and there are zero overidentifying degrees of freedom. No overidentification test can separately validate this instrument from the model alone. Exogeneity needs substantive exclusion arguments, a different source of identifying information, or additional valid restrictions. Adding instruments permits a joint consistency test of the moment conditions, but even such a test does not prove every instrument's exogeneity or identify which one is invalid without further assumptions.

3.5 Question 5: An overidentification test

Text. After 2SLS estimation of earnings on education and other covariates, with education the only endogenous variable, the output is:

$$\text{Score } \chi^2(1) = .869261, \qquad p = .3512.$$

Explain the test and its interpretation. How many instruments were used?

Complete solution. The null is that the overidentifying moment conditions are jointly valid for a correctly specified structural equation, $\mathbb{E}[z_i \varepsilon_i(\beta)] = 0$. After fitting the K coefficients using $R > K$ moments, the extra $R - K$ directions cannot all be chosen freely. A generic efficient-GMM test is

$$J = n\, g_n(\hat{\beta})' \hat{S}^{-1} g_n(\hat{\beta}) \xrightarrow{d} \chi^2_{R-K} \quad \text{under } H_0.$$

The displayed score-type test uses a related overidentification principle; it should not be relabeled as a specific homoskedastic Sargan formula merely because both have the same limiting degrees of freedom. For one degree of freedom the reported $.869261$ is below the 5% critical value 3.84146 and $p = .3512 > .05$. Thus we do not reject joint instrument compatibility with the maintained model. This is not proof that instruments are valid: the test may lack power and depends on the other structural assumptions.

The single degree of freedom means $R - K = 1$. If there are m included exogenous regressors, counting the intercept, then $K = m + 1$ because education is the one additional endogenous regressor. The instrument vector contains those m controls plus q excluded instruments, so $R = m + q$. Therefore $R - K = q - 1 = 1$, giving $\boxed{q = 2}$ excluded instruments. The total instrument count is $R = K + 1 = m + 2$; its numeric value cannot be determined without knowing the number of included controls. Distinguishing excluded instruments from the full instrument matrix avoids mistaking two for the total count.

4 Problem set 4

4.1 Question 1: Maximum likelihood estimation

Text. Consider estimating a K-dimensional parameter θ by maximizing

$$\ell_n(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log L_i(\theta).$$

(a) Describe how an estimator of the unknown parameter can be obtained.

(b) State its properties when the likelihood is correctly specified.

Complete solution.

(a) For independent observations with density or probability mass $f_i(y_i;\theta)$, the likelihood is $L(\theta) = \prod_i f_i(y_i;\theta)$, with the observations fixed and θ variable. Since the logarithm is increasing, maximizing L is equivalent to maximizing ℓ_n. Define

$$\hat{\theta} = \arg\max_{\theta\in\Theta} \ell_n(\theta).$$

For an interior differentiable optimum the score $s_n(\theta) = \partial\ell_n(\theta)/\partial\theta$ is zero. A stationary point is a local maximum if the Hessian $H_n(\theta) = \partial^2\ell_n(\theta)/\partial\theta\partial\theta'$ is negative definite there. A globally concave objective gives a global maximum; otherwise compare candidate solutions, boundary values and limiting behavior. Parameter constraints must be enforced, for example a variance must remain positive. A closed-form solution may be unavailable. Newton–Raphson iterates

$$\theta^{(m+1)} = \theta^{(m)} - H_n(\theta^{(m)})^{-1} s_n(\theta^{(m)}),$$

with line search or a constrained parameterization if needed to keep feasible values and improve the objective. Fisher scoring replaces $-H_n$ by expected information I_n, giving $\theta^{(m+1)} = \theta^{(m)} + I_n^{-1} s_n$. Verify convergence, constraints and the score/Hessian rather than assuming that any numerical output is a maximum.

(b) Correct specification alone is not sufficient for every standard property: identification, suitable support/differentiability, positive definite information, moment bounds and laws of large numbers/central limit theorems are also needed. Under those regularity conditions the estimator is consistent. One explanation is that the expected log likelihood is uniquely maximized at the true parameter θ_0: its difference from another candidate is a nonnegative Kullback–Leibler divergence, and a uniform law of large numbers transfers this population maximization to the sample.

For independent identically distributed observations define per-observation information

$$I(\theta_0) = \mathbb{E}[s_i(\theta_0)s_i(\theta_0)'] = -\mathbb{E}[H_i(\theta_0)].$$

The information equality uses the regularity conditions permitting differentiation under integration. Taylor expansion of the score around θ_0 gives

$$0 = s_n(\theta_0) + H_n(\bar{\theta})(\hat{\theta} - \theta_0),$$

so

$$\sqrt{n}(\hat{\theta} - \theta_0) = [-H_n(\bar{\theta})/n]^{-1} s_n(\theta_0)/\sqrt{n} \xrightarrow{d} N(0, I(\theta_0)^{-1}).$$

The estimator is asymptotically efficient within the regular correctly specified parametric model. Estimate its covariance by $[-H_n(\hat{\theta})]^{-1}$ or estimated expected total information $\hat{I}_n^{-1}$. MLE is not generally finite-sample unbiased or exactly normal. Its invariance property means a one-to-one reparameterization $\eta = h(\theta)$ has estimate $h(\hat{\theta})$; more generally the corresponding maximized/profile likelihood yields the transformed estimate. Under misspecification the estimator targets a pseudo-true parameter and generally needs a sandwich covariance, so the stated information-inverse formula cannot simply be assumed.

4.2 Question 2: Likelihood-based tests of linear restrictions

Text. For the preceding likelihood, consider $H_0 : R\theta = r$, with R a $J \times K$ matrix and r a $J \times 1$ vector.

(a) Describe the three testing principles and their advantages and disadvantages.

(b) Write the Wald test formula and explain it.

(c) For $K = 3$, specify R, r for $\theta_1 = \theta_2$ and $\theta_3 = 1$.

Complete solution. Assume regular interior parameter values, identification, correct likelihood specification and independent J restrictions. Let $\hat{\theta}$ maximize the likelihood without restrictions and $\tilde{\theta}$ maximize it subject to $R\theta = r$.

(a) The *likelihood-ratio* statistic is

$$LR = 2[\ell_n(\hat{\theta}) - \ell_n(\tilde{\theta})].$$

Restrictions reduce the feasible set, so $LR \geq 0$; a large likelihood loss counts against H_0. It requires both optimizations. It is invariant to one-to-one reparameterization, but can be computationally costly if either fit is difficult.

The *Wald* statistic measures how far the unrestricted estimate violates the restrictions relative to its estimated uncertainty. It requires only the unrestricted fit, convenient when the alternative is easy to estimate. Its finite-sample value can depend on how equivalent nonlinear restrictions are written, and estimated uncertainty can make it unreliable near boundaries or under weak identification.

The *score/Lagrange-multiplier* statistic measures the likelihood gradient at the restricted fit, in the directions disallowed by H_0. Using the total score and expected total information,

$$LM = s_n(\tilde{\theta})' I_n(\tilde{\theta})^{-1} s_n(\tilde{\theta}).$$

The constrained first-order conditions make the score zero in free/nuisance directions, leaving J effective directions. Equivalently parameterize $\theta = (\eta, \lambda)$ with $H_0 : \eta = \eta_0$: at the restricted fit $s_\lambda = 0$ and

$$LM = s_\eta' \left(I_{\eta\eta} - I_{\eta\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\eta} \right)^{-1} s_\eta.$$

Thus the nuisance-adjusted information is essential. The score test needs only the null fit, useful when the alternative is difficult, but depends on evaluating derivatives and information correctly at that fit.

All three converge under H_0 to χ^2_J and are first-order asymptotically equivalent under local alternatives; reject above $\chi^2_{J;1-\alpha}$. Their finite-sample values need not coincide. Boundary restrictions, nonidentification and misspecification can change these reference laws, so a blanket χ^2 claim without the regularity assumptions is invalid.

(b) Let $\hat{V}$ estimate the covariance of $\hat{\theta}$, including its $1/n$ scale. Then

$$W = (R\hat{\theta} - r)'[R\hat{V}R']^{-1}(R\hat{\theta} - r) \xrightarrow{d} \chi^2_J.$$

The covariance of the estimated restrictions is $R\hat{V}R'$. Premultiplying their deviation by its inverse measures distance in standard-error units while accounting for correlations among restrictions. If instead $\hat{A}$ estimates the covariance of $\sqrt{n}(\hat{\theta} - \theta_0)$, use $\hat{V} = \hat{A}/n$, or equivalently multiply the quadratic form using $\hat{A}$ by n. For one restriction this is the square of the asymptotic t/z statistic.

(c) Rewrite the null as $\theta_1 - \theta_2 = 0$ and $\theta_3 = 1$. The matrices are

$$R = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad r = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

They give $R\hat{\theta} - r = (\hat{\theta}_1 - \hat{\theta}_2, \hat{\theta}_3 - 1)'$ and $J = 2$. Their covariance must include the covariance between $\hat{\theta}_1$ and $\hat{\theta}_2$, rather than adding marginal standard errors.

4.3 Question 3: Logit: coefficients and marginal effects

Text. The probability of drinking is estimated as a function of gender and age. The output reports logistic regression with $n = 10843$, $LR\ \chi^2(2) = 554.51$, $\Pr(> \chi^2) = .0000$, log likelihood $= -7091.6056$ and pseudo $R^2 = .0376$.

rdrink regressor	Coefficient	SE	z	p	95% CI	
female	-.6991671	.0404696	-17.28	.000	-.778486	-.6198482
age	-.024887	.001859	-13.39	.000	-.0285306	-.0212434
constant	2.341921	.1291706	18.13	.000	2.088751	2.595091

Average marginal effects (model VCE: OIM; delta-method standard errors):

Regressor	dy/dx	SE	z	p	95% CI	
female	-.1668034	.0095985	-17.38	.000	-.1856161	-.1479906
age	-.0057471	.0004157	-13.83	.000	-.0065618	-.0049324

The female effect is the discrete change from the base (male) category.

(A) Explain the model and why it is used. Write the likelihood and log likelihood. Why not use a linear regression model?

(B) Interpret the coefficients in the first table.

(C) Interpret the second table. Give formulas for marginal effects and discuss their dependence on covariate values.

Complete solution.

(A) Let $Y_i = 1$ denote drinking and $Y_i = 0$ otherwise, with f_i the female indicator and a_i age. Conditional on covariates, use a Bernoulli law with

$$p_i = \Pr(Y_i = 1 \mid x_i) = \Lambda(x_i'\beta), \quad \Lambda(t) = \frac{e^t}{1+e^t}, \quad x_i = (1, f_i, a_i)'.$$

For conditionally independent observations,

$$L(\beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i},$$

and

$$\begin{aligned}\ell(\beta) &= \sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)] \\ &= \sum_i [y_i x_i'\beta - \log(1+e^{x_i'\beta})].\end{aligned}$$

Differentiating gives $\sum_i x_i(y_i - p_i)$, and the Hessian is $-\sum_i p_i(1-p_i)x_i x_i'$. The estimates solve these nonlinear score equations. The logit link restricts predictions to $(0,1)$ and permits probability changes to vary with covariates. A linear probability model $\mathbb{E}[Y \mid x] = x'\beta$ can be estimated but may predict outside $[0,1]$ and has heteroskedastic conditional variance $p_i(1-p_i)$; it also imposes constant slopes on probabilities. These are reasons to prefer a suitably specified binary-response model, not a claim that linear regression of an indicator is mathematically impossible.

(B) The log odds satisfy $\log[p_i/(1-p_i)] = x_i'\beta$. Thus a one-year age increase, holding gender fixed, reduces fitted log odds by .024887, multiplying the odds by $e^{-.024887} \approx .97542$, a 2.458% reduction. The female

coefficient compares otherwise identical women with men: their odds ratio is $e^{-.6991671} \approx .496999$, about 50.30% lower odds. The negative signs imply lower probabilities because Λ is increasing, but the coefficient magnitudes are not probability changes. The constant 2.341921 is the male log odds at age zero, which is an extrapolated baseline rather than a typical adult's odds. The individual z statistics show strong evidence against zero coefficients under the maintained asymptotic likelihood model; statistical association is not automatically causation.

(C) For a continuous variable x_k, differentiate the link:

$$\frac{\partial p_i}{\partial x_{ik}} = \Lambda'(x_i'\beta)\beta_k = p_i(1-p_i)\beta_k.$$

Since p_i depends on all covariates, this derivative differs across individuals even though β_k is constant. The age average marginal effect is $n^{-1}\sum_i \hat{p}_i(1-\hat{p}_i)\hat{\beta}_{age} = -.0057471$, roughly $-.57471$ percentage points per year on average. This derivative is a local change; an exact one-year increment is $\Lambda(\eta_i+\beta_{age})-\Lambda(\eta_i)$.

For a binary regressor differentiation is not the right comparison. Set female first to one then to zero for each individual at the same age:

$$\Delta_i = \Lambda(\beta_0 + \beta_f + \beta_a a_i) - \Lambda(\beta_0 + \beta_a a_i).$$

Its sample average $-.1668034$ means a fitted drinking-probability difference of -16.68034 percentage points for women, averaged over observed ages. It is not a 16.68% relative probability decline. The average of individual derivatives generally differs from the derivative evaluated at average covariates. Delta-method errors propagate coefficient uncertainty through the derivative of these effect formulas; the reported 95% intervals quantify that uncertainty. The pseudo R^2 is a likelihood-based fit measure, not an OLS fraction of variance explained.

4.4 Question 4: A model for labor-force participation

Text. We want to study how the probability of choosing to work or not work varies with individual characteristics. Sketch an appropriate model and its interpretation; is it appropriate to interpret the coefficients directly?

Complete solution. Define $Y_i = 1$ if an individual works and $Y_i = 0$ otherwise. One model is $\Pr(Y_i = 1 \mid x_i) = \Lambda(x_i'\beta)$ with the logistic link $\Lambda(t) = e^t/(1+e^t)$. The conditional Bernoulli likelihood is $\prod_i p_i^{y_i}(1-p_i)^{1-y_i}$ and is maximized over β. An alternative probit link is $\Phi(x_i'\beta)$, where Φ is the standard normal cumulative distribution. A latent-index description is $Y_i = \mathbf{1}\{x_i'\beta + v_i > 0\}$, with logistic or standard normal v_i and suitable conditional independence/exogeneity assumptions. The index scale is normalized because binary outcomes do not identify its arbitrary absolute scale.

In a logit, coefficients are changes in log odds; in a probit they are changes in the normalized latent index. Neither is directly a change in the probability of working. For a continuous regressor the probability derivative is $\beta_k\Lambda(\eta_i)(1-\Lambda(\eta_i))$ in logit and $\beta_k\phi(\eta_i)$ in probit, with $\eta_i = x_i'\beta$ and ϕ the standard normal density. For a dummy compute the difference in predicted probability when switching it from zero to one, holding other variables fixed. Report these effects at specified covariates or average them over the sample, with standard errors or confidence intervals. Interpret fitted associations as causal only if the identification and exogeneity assumptions justify that stronger claim.

5 Problem set 5

5.1 Question 1: Minimum wages in a state panel

Text. A study of minimum wages and teenage employment uses annual observations for 1977–1989 on 50 US states and the District of Columbia. The equation is

$$E_{it} = \beta_0 + \beta_1(M_{it}/W_{it}) + \sum_{j=2}^{51} \gamma_j D_{ji} + \sum_{s=2}^{13} \delta_s B_{st} + u_{it},$$

with other covariates also included. Here E is the teenage employment-to-population ratio, M the nominal minimum wage, W the state's average wage, D_{ji} state dummies and B_{st} time dummies. The other covariates are the teenage population share SHY and the prime-age male unemployment rate uram.

(a) Discuss the advantage of this panel over pure cross-sectional or time-series data.

(b) With only time dummies included, the estimates are

$$\widehat{E}_{it} = \hat{\beta}_0 - .33(M_{it}/W_{it}) + .35SHY_{it} - 1.53uram_{it}, \quad R^2 = .20,$$

with standard errors $.08, .28, .13$ respectively. Time coefficients are omitted from the display. Comment on significance and explain how to calculate elasticities in a level regression.

(c) Adding state effects gives coefficients $.07, -.19, -.54$, with standard errors $.10, .22, .11$ and $R^2 = .69$. Compare the results and explain the change.

Complete solution.

(a) A panel follows the same states across time, so state effects can absorb time-invariant omitted factors such as geography, persistent industrial composition or institutions. Year effects absorb shocks common to all states, such as national macroeconomic conditions. Identification of the slope after both sets of effects relies on remaining within-state changes net of common year changes, rather than merely comparing permanently different states. Repeated observations also increase the usable information, but they are not automatically independent: common and serially correlated shocks must be reflected in inference. State effects cannot remove omitted factors that change within states in ways correlated with policy.

(b) Using the reported errors, the three test statistics for zero slopes are

$$t_M = -.33/.08 = -4.125, \quad t_{SHY} = .35/.28 = 1.25, \quad t_{uram} = -1.53/.13 \approx -11.7692.$$

With the usual large-sample two-sided cutoffs 1.96 at 5% and 2.576 at 1%, the minimum-wage ratio and unemployment slopes are significant at 1%; the teenage-share slope is not significant at 5%. Holding other variables and year effects fixed, a .01 increase in M/W lowers the fitted employment ratio by .0033. If E is a proportion, this equals .33 percentage points. The .20 fit measure describes explained in-sample variation; it does not establish causality.

For a level equation $m(x) = \beta_0 + \beta_x x + \cdots$, the elasticity at a specified covariate vector is

$$\epsilon_x = \frac{\partial m}{\partial x}\frac{x}{m(x)} = \beta_x \frac{x}{m(x)}.$$

It varies with x and the fitted outcome. Evaluating it at sample means uses $\hat{\beta}_x \bar{x}/\widehat{m}(\bar{x})$ (or a stated representative E); a numerical value cannot be obtained without those levels. For the minimum wage M itself holding W fixed, $\partial E/\partial M = \beta_1/W$, so its elasticity is $\beta_1(M/W)/E$. A coefficient of $-.33$ in a level-ratio model is not itself an elasticity.

(c) With state effects, $t_M = .07/.10 = .7$, $t_{SHY} = -.19/.22 \approx -.864$, and $t_{uram} = -.54/.11 \approx -4.909$. The policy coefficient changes sign and becomes statistically indistinguishable from zero; the teenage-share coefficient also changes sign and remains insignificant. The unemployment association remains negative and significant but its magnitude is smaller. The fitted R^2 rises to .69 because added state indicators explain persistent differences; such an in-sample increase alone is not a model-validity test.

The pooled/time-only model may have omitted state characteristics correlated with both regressors and employment. State indicators change the comparisons used to estimate slopes and can remove that source of omitted-variable bias. This is a plausible explanation of the difference, not proof that every remaining bias is removed. The supplied standard errors are homoskedastic-only. With state-level repeated observations, within-state dependence can make them unreliable; the stated significance calculations are conditional on using those supplied errors, while valid panel inference generally needs an appropriate covariance estimate, often clustering by state.

5.2 Question 2: An intercept after differencing

Text. For 1982 and 1988 consider

$$FatalityRate_{i,1982} = \beta_0 + \beta_1 BeerTax_{i,1982} + u_{i,1982},$$
$$FatalityRate_{i,1988} = \beta_0 + \beta_1 BeerTax_{i,1988} + u_{i,1988}.$$

FatalityRate measures annual traffic deaths per 10000 people, and BeerTax is the tax on a case of beer in 1988 US dollars. After subtracting the first equation from the second, a negative intercept is estimated.

(a) Modify the equations to allow an intercept in the differenced model.

(b) What relative magnitude must the modified intercepts have for a negative differenced intercept?

Complete solution.

(a) With exactly the same intercept β_0 in both equations, subtraction cancels it: $\Delta FatalityRate_i = \beta_1 \Delta BeerTax_i + \Delta u_i$. To permit a common change over time unrelated to the tax, assign period-specific intercepts α_{82} and α_{88}:

$$FatalityRate_{i,t} = \alpha_t + \beta_1 BeerTax_{i,t} + u_{i,t}, \qquad t \in \{1982, 1988\}.$$

Subtracting gives

$$\Delta FatalityRate_i = (\alpha_{88} - \alpha_{82}) + \beta_1 \Delta BeerTax_i + \Delta u_i.$$

Equivalently use a base intercept and a 1988 dummy with coefficient δ, so the differenced intercept equals δ. If a time-invariant state effect is also included, it cancels under differencing and does not prevent this time intercept.

(b) The population differenced intercept is negative iff $\alpha_{88} < \alpha_{82}$. It represents a downward common time shift in fatality rates, holding tax changes fixed. A negative sample estimate supports that interpretation within its uncertainty; statistical significance would require its standard error, not supplied here.

5.3 Question 3: Unobserved heterogeneity and identification

Text. Consider one cross-section of $y_{i1} = \beta x_{i1} + \alpha_i + u_{i1}$ with $\mathbb{E}(u_{i1} \mid x_{i1}, \alpha_i) = 0$.

(a) When can β be consistently estimated with only this cross-section?

(b) If $T = 2$ and the same individuals are observed at both dates but α_i is unobserved, how can β be identified?

(c) What correlation assumptions between x and α_i does the proposed panel estimator need?

Complete solution.

(a) If α_i is observed and not collinear with x, include it as a regressor and estimate the structural slope under the given exogeneity assumption and ordinary rank/moment conditions. If it is unobserved, absorb its mean in an intercept and write $\alpha_i = \mathbb{E}\alpha_i + a_i$. The OLS population slope is

$$\beta + \frac{\mathrm{Cov}(x_{i1}, a_i)}{\mathrm{Var}(x_{i1})},$$

because the conditional-zero-mean assumption implies $\mathrm{Cov}(x_{i1}, u_{i1}) = 0$. Thus nonzero regressor variance and $\mathrm{Cov}(x_{i1}, \alpha_i) = 0$ permit consistent OLS. If that orthogonality fails, a valid excluded instrument z_i can identify β provided $\mathbb{E}[z_i(\alpha_i + u_{i1})] = 0$ after centering/controls and $\mathbb{E}[z_i x_{i1}] \neq 0$. The supplied conditional error mean alone does not make an unobserved α_i orthogonal to x.

(b) Assuming the same slope and individual effect at both dates, subtract the two equations:

$$\Delta y_i = \beta \Delta x_i + \Delta u_i,$$

so the unobserved effect cancels. The first-difference estimate is

$$\hat{\beta}_{FD} = \frac{\sum_i \Delta x_i \Delta y_i}{\sum_i (\Delta x_i)^2}.$$

Identification and consistency require $\mathbb{E}[(\Delta x_i)^2] > 0$ and $\mathbb{E}[\Delta x_i \Delta u_i] = 0$, with finite moments and a law of large numbers across individuals. If a common time shift is included, difference it into an intercept and use centered differences for the slope.

(c) The first-difference estimator permits arbitrary correlation between α_i and either date's regressors: the effect is removed algebraically. It does need exogeneity relative to the differenced idiosyncratic error. A sufficient condition is strict exogeneity

$$\mathbb{E}[u_{it} \mid x_{i1}, x_{i2}, \alpha_i] = 0, \qquad t = 1, 2.$$

The weaker contemporaneous conditions $\mathbb{E}[u_{it} \mid x_{it}, \alpha_i] = 0$ alone do not rule out correlations such as x_{i2} with u_{i1}, and hence do not guarantee $\mathbb{E}[\Delta x_i \Delta u_i] = 0$. Eliminating fixed heterogeneity does not eliminate this feedback/endogeneity problem.

5.4 Question 4: Fixed effects: LSDV, within and first differences

Text.

1) Write the linear fixed-effects model and its assumptions.

2) Derive the least-squares dummy-variable (LSDV) estimate and discuss its properties.

3) Derive the within estimator and discuss its properties.

4) Derive the first-difference estimator and discuss its properties.

Complete solution.

1) For individuals $i = 1, \ldots, N$ and dates $t = 1, \ldots, T$ use

$$y_{it} = \alpha_i + x_{it}'\beta + u_{it}.$$

The common slope β is K dimensional, while the individual intercepts α_i may correlate arbitrarily with the full regressor history. For the standard unbiased/consistent slope results assume strict exogeneity

$\mathbb{E}[u_{it} \mid x_{i1}, \ldots, x_{iT}, \alpha_i] = 0$, sufficient moments, independent individuals or an appropriate cross-sectional law of large numbers, and full rank of the within-regressor moment matrix. Regressors must have within-individual variation: a time-invariant variable is collinear with individual effects and its separate slope is not identified. Common time effects may be added with a nonredundant dummy normalization. Homoskedasticity and no serial correlation are additional efficiency/inference assumptions, not prerequisites for removal of α_i.

2) Stack the observations as $y = D\alpha + X\beta + u$, where D is the $NT \times N$ matrix of individual dummies. Include all $dummies$ without an extra constant, or a constant with one omitted dummy, to avoid collinearity. The normal equation for α gives

$$\hat{\alpha} = (D'D)^{-1}D'(y - X\hat{\beta}).$$

Substitution into the normal equation for β, with $M_D = I - D(D'D)^{-1}D'$, yields

$$\hat{\beta}_{LSDV} = (X'M_DX)^{-1}X'M_Dy.$$

Since $M_DD = 0$, $\hat{\beta}_{LSDV} - \beta = (X'M_DX)^{-1}X'M_Du$. Strict exogeneity makes the conditional error mean zero, giving conditional unbiasedness where moments exist. As $N \to \infty$ with fixed T, a full-rank within moment matrix and suitable laws of large numbers give consistency of $\hat{\beta}$. Individual intercepts estimated with a fixed number of observations per individual need not themselves be consistent as N grows. This does not invalidate the linear fixed-effect slope result. With $\mathrm{Var}(u \mid X, D) = \sigma^2 I$, the slope covariance is $\sigma^2(X'M_DX)^{-1}$; residual degrees of freedom are $NT - N - K$ for this specification.

3) For a balanced panel define $\bar{y}_i = T^{-1}\sum_t y_{it}$, $\bar{x}_i = T^{-1}\sum_t x_{it}$, $\bar{u}_i = T^{-1}\sum_t u_{it}$. Average and subtract:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (u_{it} - \bar{u}_i).$$

The individual effect disappears because it is unchanged across dates. Write $\tilde{y}_{it} = y_{it} - \bar{y}_i$ and $\tilde{x}_{it} = x_{it} - \bar{x}_i$. OLS in the transformed equation gives

$$\hat{\beta}_W = \left(\sum_{i,t} \tilde{x}_{it}\tilde{x}'_{it}\right)^{-1} \sum_{i,t} \tilde{x}_{it}\tilde{y}_{it}.$$

The demeaning operation is precisely M_D, so this is algebraically the LSDV slope, with identical fitted slopes and residuals. Strict exogeneity verifies orthogonality of $\tilde{x}_{it}$ to $\tilde{u}_{it}$, including the terms involving individual means; merely contemporaneous exogeneity is insufficient. The same rank, unbiasedness and consistency properties apply. The transformed errors are correlated across dates even if the original errors were independent, so inference must respect the transformation and actual panel covariance. If serial correlation or heteroskedasticity is permitted, one cluster-robust slope covariance is

$$\widehat{V} = Q^{-1}\left[\sum_i \left(\sum_t \tilde{x}_{it}\hat{u}_{it}\right)\left(\sum_t \tilde{x}_{it}\hat{u}_{it}\right)'\right]Q^{-1}, \quad Q = \sum_{i,t} \tilde{x}_{it}\tilde{x}'_{it},$$

under suitable asymptotics with many independent clusters.

4) For $t = 2, \ldots, T$, subtract the preceding observation:

$$\Delta y_{it} = (\Delta x_{it})'\beta + \Delta u_{it}.$$

The estimator is

$$\hat{\beta}_{FD} = \left(\sum_{i,t\geq 2} \Delta x_{it}\Delta x'_{it}\right)^{-1} \sum_{i,t\geq 2} \Delta x_{it}\Delta y_{it}.$$

It is consistent if the differenced regressor moment matrix is full rank and $\mathbb{E}[\Delta x_{it}\Delta u_{it}] = 0$, together with appropriate moments and a law of large numbers. Strict exogeneity supplies this condition and conditional

unbiasedness, but weaker special conditions can suffice. Time-invariant regressors again disappear. For $T > 2$, first differences and within estimates need not coincide. If original u_{it} are independent with variance σ^2, then $\mathrm{Var}(\Delta u_{it}) = 2\sigma^2$ and $\mathrm{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) = -\sigma^2$. Consequently treating differences as independent can invalidate ordinary errors; use the appropriate GLS or cluster covariance. Relative efficiency depends on the original error process: within is efficient under spherical original errors, while differencing may be attractive when original errors are strongly persistent. Neither transformation by itself resolves time-varying omitted variables or endogenous regressors.

5.5 Question 5: Equality of FE and FD when $T = 2$

Text. Consider $y_{it} = \beta x_{it} + \alpha_i + u_{it}$ with $\mathbb{E}(u_{it} \mid x_{it}, \alpha_i) = 0$ and $T = 2$. Show that fixed effects and first differences give the same slope estimator.

Complete solution. Define $\Delta y_i = y_{i2} - y_{i1}$ and $\Delta x_i = x_{i2} - x_{i1}$. Subtracting equations removes α_i, so the FD slope is

$$\hat{\beta}_{FD} = \frac{\sum_i \Delta x_i \Delta y_i}{\sum_i (\Delta x_i)^2}.$$

For fixed effects,$\bar{y}_i = (y_{i1} + y_{i2})/2$ and $\bar{x}_i = (x_{i1} + x_{i2})/2$. Therefore

$$\begin{aligned} y_{i1} - \bar{y}_i &= -\tfrac{1}{2}\Delta y_i, & y_{i2} - \bar{y}_i &= \tfrac{1}{2}\Delta y_i, \\ x_{i1} - \bar{x}_i &= -\tfrac{1}{2}\Delta x_i, & x_{i2} - \bar{x}_i &= \tfrac{1}{2}\Delta x_i. \end{aligned}$$

For each individual, the within numerator contribution is

$$(-\tfrac{1}{2}\Delta x_i)(-\tfrac{1}{2}\Delta y_i) + (\tfrac{1}{2}\Delta x_i)(\tfrac{1}{2}\Delta y_i) = \tfrac{1}{2}\Delta x_i \Delta y_i.$$

The corresponding denominator contribution is $\frac{1}{2}(\Delta x_i)^2$. Summing across individuals cancels the common factor:

$$\hat{\beta}_{FE} = \frac{\sum_i \frac{1}{2}\Delta x_i \Delta y_i}{\sum_i \frac{1}{2}(\Delta x_i)^2} = \boxed{\hat{\beta}_{FD}}.$$

The required denominator is positive, meaning at least one individual has a regressor change. This is an algebraic identity for the stated model, requiring no probabilistic exogeneity assumption. Consistency of their common slope additionally needs $\mathbb{E}[\Delta x_i \Delta u_i] = 0$; the displayed contemporaneous condition alone need not imply it. With several regressors the same factor cancels from the cross-product matrices. Matching time effects and other controls must be included in both transformations for the equivalent model comparison.